Prediction of PVT Behaviour and Thermodynamic Properties (H and S) of Real Gases Using Cubic Equations of State

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1. Objective

The project objective is to develop software that predicts the pressure–volume–temperature (PVT) behaviour of real gases with a cubic equation of state (EOS). The follow-up assignment extends the program to compute the thermodynamic properties enthalpy (H) and entropy (S). The program is written in Python (NumPy/SciPy) and contains four cubic EOS (van der Waals, Redlich–Kwong, Soave–Redlich–Kwong and Peng–Robinson), a small data base of six gases, and an interactive calculator.

2. Theory

2.1 Generalised cubic EOS. All four models are cast in the single form of Smith, Van Ness and Abbott [1]:

$$P = RT/(V - b) - a(T) / [(V + \varepsilon b)(V + \sigma b)] \quad (1)$$

with $b = \Omega\, RTc/Pc$ and $a(T) = \Psi\, \alpha(Tr,\omega)\, R^2Tc^2/Pc$. The constants are given in Table 1. Only α, σ, ε, Ω and Ψ change from one EOS to another, so a single code path serves all models.

EOS	σ	ε	Ω	Ψ	α(Tr, ω)	d ln α / d ln Tr
vdW	0	0	1/8	27/64	1	0
RK	1	0	0.08664	0.42748	$Tr^{-0.5}$	−0.5
SRK	1	0	0.08664	0.42748	$[1 + \kappa(1 - \sqrt{Tr})]^2$, $\kappa = 0.480 + 1.574\omega - 0.176\omega^2$	$-\kappa\sqrt{Tr} / [1 + \kappa(1 - \sqrt{Tr})]$
PR	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	$[1 + \kappa(1 - \sqrt{Tr})]^2$, $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$	same as SRK

Table 1. Parameters of the generalised cubic equation of state.

2.2 Compressibility factor. With $\beta = \Omega Pr/Tr$ and $q = \Psi\alpha/(\Omega Tr)$, the EOS becomes a cubic polynomial in $Z = PV/RT$:

$$Z^3 + [(\varepsilon+\sigma)\beta - (1+\beta)]Z^2 + [\varepsilon\sigma\beta^2 - (1+\beta)(\varepsilon+\sigma)\beta + q\beta]Z - [(1+\beta)\varepsilon\sigma\beta^2 + q\beta^2] = 0 \quad (2)$$

The program finds the roots with numpy.roots and keeps the real roots with $Z > \beta$. Three roots occur below Tc inside the two-phase region: the largest is the vapour root and the smallest is the liquid root. The vapour pressure is found from the equal-fugacity condition $\ln\varphi(\text{liquid}) = \ln\varphi(\text{vapour})$, where $\ln\varphi = Z - 1 - \ln(Z - \beta) - qI$ and $I = \ln[(Z+\sigma\beta)/(Z+\varepsilon\beta)]/(\sigma-\varepsilon)$ ($I = \beta/Z$ for vdW). The solution is bracketed and solved with Brent's method, and the result decides which root is stable at a given P.

2.3 Enthalpy and entropy (HW). Each property is the sum of an ideal-gas part and a residual (departure) part. The residual parts follow from the EOS [1]:

$$H^R/RT = Z - 1 + (d\ln\alpha/d\ln Tr - 1)\, q\, I \quad (3)$$

$$S^R/R = \ln(Z - \beta) + (d\ln\alpha/d\ln Tr)\, q\, I \quad (4)$$

For vdW the term d ln α/d ln Tr is zero. The ideal-gas parts are obtained by integrating the heat capacity $Cp^{ig}/R = A + BT + D/T^2$, with the reference state of an ideal gas at T0 = 298.15 K and P0 = 1 bar (H = S = 0):

$$H = R\int Cp^{ig}/R\, dT + H^R\,, \quad S = R\int (Cp^{ig}/R)/T\, dT - R\, ln(P/P0) + S^R \tag{5}$$

The integrals are evaluated analytically: $\int Cp/R\, dT = AT + BT^2/2 - D/T$ and $\int (Cp/R)/T\, dT = A \ln T + BT - D/(2T^2)$. The same formulas hold for the liquid root, so the enthalpy of vaporisation is $\Delta Hvap = H^R(\text{vapour}) - H^R(\text{liquid})$ at Psat.

3. Program description

The code (real_gas_eos.py) is organised into small functions that follow the theory: (i) a gas data base with Tc, Pc, ω and Cp^{ig} constants (methane, ethane, propane, n-butane, CO2, N2) [1]; (ii) EOS definitions; (iii) z_roots() for Eq. (2); (iv) psat() for vapour pressure; (v) departure() and properties() for Eqs. (3)–(5); and (vi) an interactive front end. The user selects a gas and an EOS and enters T [K] and P [bar]; the program prints the phase, Z, molar volume, H^R, S^R, H and S. Running the program with the --demo option reproduces every table and figure of this report.

4. Results and discussion

4.1 Comparison of the EOS. Table 2 compares the four models for propane at 400 K and 20 bar (Tr = 1.08, Pr = 0.47), where the gas is slightly non-ideal. All models give Z < 1 because intermolecular attraction dominates at this state. The residual terms of RK, SRK and PR agree within about 15%, while vdW gives noticeably smaller departures, a consequence of its simple attractive term.

EOS	Z	V [cm³/mol]	H^R [J/mol]	S^R [J/(mol K)]	H [J/mol]	S [J/(mol K)]
vdW	0.8718	1449.8	−1073.8	−1.676	8462.4	0.756
RK	0.8619	1433.2	−1396.3	−2.388	8140.0	0.043
SRK	0.8659	1439.9	−1523.4	−2.735	8012.8	−0.304
PR	0.8503	1414.0	−1581.3	−2.745	7954.9	−0.314

Table 2. Propane at 400 K and 20 bar (reference: ideal gas at 298.15 K, 1 bar).

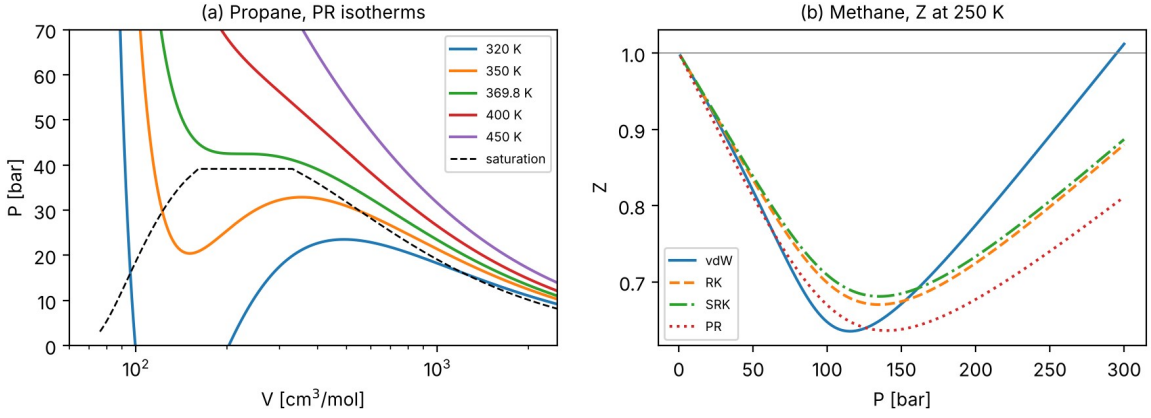


Figure 1. (a) PR isotherms for propane with the saturation envelope; (b) compressibility factor of methane at 250 K from the four EOS.

Figure 1a shows the typical cubic behaviour: below Tc = 369.8 K the isotherms contain a van der Waals loop (three roots) that is replaced by the horizontal tie line at Psat, the 369.8 K isotherm shows the inflection point, and above Tc the isotherms decrease monotonically. In Figure 1b, Z of methane falls below 1 as attractive forces dominate, reaches a minimum and then rises as repulsion takes over. vdW predicts the minimum at too low a pressure and a Z that is too high at 300 bar, so it is only qualitatively correct, whereas PR gives the lowest Z. The critical compressibility factors of the models are 0.375 (vdW), 0.333 (RK, SRK) and 0.307 (PR), compared with 0.27–0.29 for real fluids, which explains why cubic EOS overestimate liquid molar volumes (PR least).

4.2 Enthalpy and entropy. Figure 2 shows the PR residual properties of methane. H^R and S^R are negative, vanish as $P \rightarrow 0$ (ideal gas), and increase in magnitude with rising pressure and falling temperature, because the stronger molecular interactions order the fluid and lower its enthalpy and entropy relative to the ideal gas. The 200 K isotherm (just above Tc = 190.6 K) changes steeply near 55–65 bar where the fluid densifies rapidly, while the 500 K isotherm is almost linear.

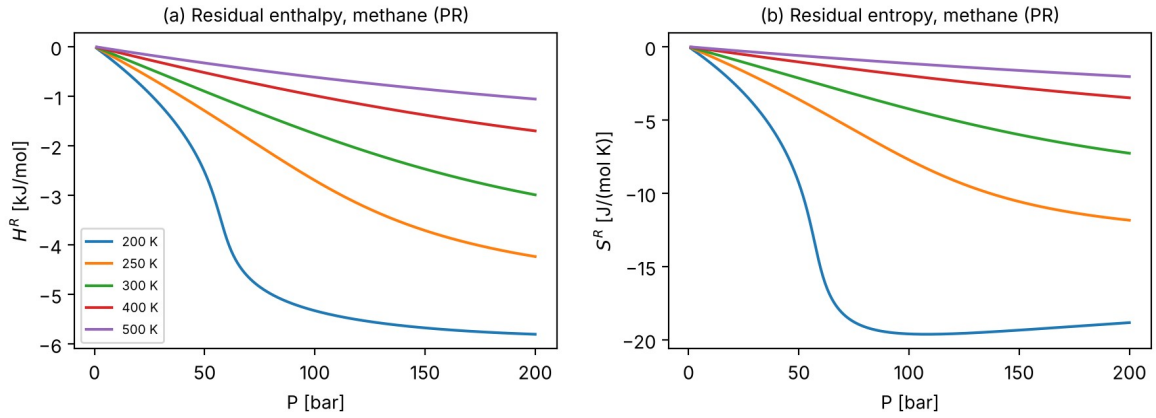


Figure 2. Residual enthalpy and entropy of methane from the PR EOS at several temperatures.

4.3 Validation. Two checks were made. First, the enthalpy of vaporisation at the normal boiling point was computed from Eq. (3) at the EOS vapour pressure and compared with handbook values (Table 3). PR agrees within 0.4% and SRK within 3%. The predicted Psat is also close to 1.013 bar (0.98–1.02 bar). Second, the thermodynamic identity $(\partial H/\partial P)T = V - T(\partial V/\partial T)P$ was tested numerically for methane (PR, 300 K, 50 bar): both sides equal −179.07 cm³/mol, which confirms that Eq. (3) and the Z-roots are mutually consistent.

Gas	Tb [K]	ΔHvap lit. [kJ/mol]	ΔHvap SRK [kJ/mol]	Error SRK	ΔHvap PR [kJ/mol]	Error PR
Methane	111.67	8.19	8.43	+2.9%	8.21	+0.3%
Nitrogen	77.36	5.56	5.67	+2.0%	5.54	−0.3%
Propane	231.04	18.77	19.04	+1.4%	18.69	−0.4%

Table 3. Enthalpy of vaporisation at the normal boiling point.

5. Conclusions

- The program solves the cubic EOS for vdW, RK, SRK and PR, selects the correct phase root through the vapour pressure, and returns Z, V, H and S for six gases.
- Residual enthalpy and entropy are obtained analytically from the EOS and combined with ideal-gas integrals, and the results are thermodynamically consistent.
- PR and SRK are the most accurate models; PR is best for ΔHvap (error < 0.5%), while vdW is only qualitative. Limitations: pure components only, and Cp^{ig} polynomials valid at roughly 300–1500 K, so absolute H and S far outside this range are approximate. Extension to mixtures through mixing rules is the natural next step.

References

[1] J. M. Smith, H. C. Van Ness, M. M. Abbott, Introduction to Chemical Engineering Thermodynamics, McGraw-Hill (Ch. 3, 6; Tables B.1, C.1).

[2] D.-Y. Peng, D. B. Robinson, Ind. Eng. Chem. Fundam. 15 (1976) 59–64. [3] G. Soave, Chem. Eng. Sci. 27 (1972) 1197–1203.

[4] NIST Chemistry WebBook (normal boiling points and enthalpies of vaporisation).